

1. Administrative & Study Identification

Full Study Title: An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of Efinopegdutide (MK-6024) in Participants With Moderate and Severe Hepatic Impairment
Short Title/Acronym: A Study of Efinopegdutide in Participants With Hepatic Impairment (MK-6024-014)
Protocol Number: MK-6024-014
NCT Number: NCT06052566
Sponsor Name: Merck Sharp & Dohme LLC
Investigational Product: Efinopegdutide (also known as MK-6024, HM12525A, JNJ-64565111)
Phase of Development: Phase 1
Medical Monitor: Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (PK) of efinopegdutide in participants with hepatic impairment compared to healthy participants.
Secondary Objectives: To examine the safety and tolerability of a single dose of efinopegdutide in participants with moderate and severe hepatic impairment.

2.2 Background & Rationale Efinopegdutide is an investigational drug being studied for the treatment of liver conditions such as metabolic dysfunction-associated steatohepatitis (MASH/NASH). Because patients with these conditions often present with varying degrees of liver damage, it is crucial to understand how hepatic impairment affects the drug's metabolism and clearance. This study provides necessary PK and safety data to guide dosing for patients with underlying moderate or severe hepatic impairment.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional
Control Type: Parallel Assignment (Matched healthy control participants vs. hepatic impaired participants)
Blinding: Open-label
Randomization: N/A

3.2 Planned Enrollment & Duration Target \$N\$: 22 participants
Number of Sites: United States
Estimated Study Start: November 21, 2023
Estimated Primary Completion: December 5, 2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. A participant assigned female at birth is eligible to participate if not pregnant or breastfeeding, is not a participant of childbearing potential (POCBP), or is a POCPB and agrees to follow contraceptive guidance during the study intervention period and for at least 35 days after the last dose.

2. For participants with moderate or severe hepatic impairment: Have a diagnosis of chronic (>6 months), stable, hepatic impairment with features of cirrhosis due to any etiology.

3. Stability of hepatic disease should correspond to no acute episodes of illness within the previous 2 months due to deterioration in hepatic function.

4.2 Exclusion Criteria 1. History of cancer (malignancy).

2. Had major surgery and/or donated or lost 1 unit of blood (approximately 500 mL) within 4 weeks prior to the prestudy (screening) visit.

3. Participants with moderate or severe hepatic impairment who are positive for human immunodeficiency virus (HIV)-1 or HIV-2 at the screening visit.

4. Received antiviral and/or immune-modulating therapy for hepatitis B virus (HBV) or hepatitis C virus (HCV) within 90 days prior to study start.

5. Healthy participants unable to refrain from the use of any medication (prescription/nonprescription/herbal) beginning approximately 2 weeks (or 5 half-lives) prior to initial dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single 7 mg dose of efinopegdutide (14 mg/mL).

Route of Administration: Subcutaneous (SC) injection.

Duration of Treatment: Single-dose administration on Day 1.

5.2 Concomitant & Prohibited Therapy Permitted: Stable baseline treatments for hepatic impairment and related chronic co-morbidities.

Prohibited: Antiviral or immune-modulating therapies for HBV/HCV within 90 days prior to study start; any unauthorized prescription or herbal medications for healthy controls 14 days before dosing.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to -28 Days Informed Consent, Medical History, Hepatic Assessment, HIV/HBV/HCV Labs
Baseline	Day 1	Single SC injection of Efinopegdutide (7 mg)	
Interim PK Visits	Days 1-8, 11, 14, 21	PK Blood Sampling,	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Area Under the Curve From Time 0 to Infinity (AUC_{0-inf}) of Efinopegdutide.
Measurement Method: Blood samples collected at pre-specified timepoints post-dose on Days 1 through 35.

7.2 Secondary & Exploratory Endpoints

1. Safety and Tolerability (number of participants experiencing clinical adverse events).
2. Maximum Observed Concentration (C_{max}) of Efinopegdutide.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation during clinical visits through Day 35 post-dose.
Serious Adverse Events (SAEs): Reported within standard 24-hour safety timelines to the Sponsor.
Data Monitoring Committee (DMC): Internal safety and pharmacovigilance team overseen by Merck Sharp & Dohme LLC.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving \$\\ge 1\$ dose of efinopegdutide); PK Analysis Set.
Sample Size Justification: \$N=22\$ participants is adequate for a Phase 1 PK evaluation to detect significant variations in drug metabolism across hepatic impairment severities.
Handling of Missing Data: Standard non-compartmental pharmacokinetic analysis techniques.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Standard clinical site monitoring to ensure protocol adherence and safety compliance.
Audit Trail: Validated eCRF tracking and laboratory PK sample tracking logs.

11. Study Identifiers & Governance

Primary ID: {{MK-6024-014}}
Registry Number: {{NCT06052566}}
Sponsor/Lead Organization: {{Merck Sharp & Dohme LLC}}

Study Title: {{A Study of Efinopegdutide in Participants With Hepatic Impairment (MK-6024-014)}}
Official Regulatory Title: {{An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of Efinopegdutide (MK-6024) in Participants With Moderate and Severe Hepatic Impairment}}

12. Clinical Framework (Hepatic Impairment Specific)

Primary Condition: {{Hepatic Impairment / Metabolic Dysfunction Associated Steatohepatitis}}
Diagnosis Threshold: {{Chronic (>6 months), stable, hepatic impairment with features of cirrhosis}}
Medical Methodology: {{Pharmacokinetic (PK) Evaluation}}
Optimization Target: {{Safety, Tolerability, and PK profile mapping}}

13. Study Procedures & Methodology

Management Pathway: {{Standard Phase 1 PK and Safety Monitoring}}
Education Protocol (HCP): {{Study protocol training and AE reporting procedures}}
Education Protocol (Patient): {{Informed consent and protocol requirement review}}
Quality Audit Mechanism: {{Clinical site monitoring and PK sample tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{35 Days post-dose}}
Onsite Visit Cadence: {{Days 1-8, 11, 14, 21, and 35}}
Remote Monitoring Frequency: {{As needed for safety}}
Checklist Review Interval: {{Per protocol schedule}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				